

BST Rotation Curves: SPARC Calculation

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Reproducible code and results for the BST channel noise rotation curve calculation against the full SPARC database (175 galaxies, zero free parameters).

The theoretical basis is documented in the BST Working Paper (root directory). The complete derivation and analysis are in `DarkMatterCalculation.md` (available on request; PDF forthcoming).

Quick Start

Run all 175 SPARC galaxies

```
python3 sparc_bst.py
```

Single galaxy

```
python3 sparc_bst.py --galaxy NGC3198
```

Quality-1 galaxies only

```
python3 sparc_bst.py --quality 1
```

Use BST-derived $a_0 = c^2/2\pi R_H$ instead of observed

```
python3 sparc_bst.py --a0-bst
```

Requires Python 3, numpy, scipy.

Files

File	Description
<code>sparc_bst.py</code>	Main calculator — reads SPARC rotmod files, applies BST acceleration formula, outputs per-galaxy metrics
<code>SPARC_BST_Results.csv</code>	Full results: RMS, TF error, dark non-matter fractions, reduced χ^2 for all 175 SPARC galaxies
<code>SPARC_data/</code>	SPARC rotation curve files (Lelli, McGaugh & Schombert 2016) — see attribution below

The BST Formula

Dark non-matter = incomplete windings on the S^1 channel (channel noise). The total acceleration satisfies:

$$\mu(a_{\text{total}} / a_0) \cdot a_{\text{total}} = a_{\text{Newton}}$$

$$\mu(x) = x / \sqrt{1 + x^2} \quad \leftarrow \text{derived from Shannon S/N of Haldane channel}$$

$$a_0 = c^2 / 2\pi R_H = 1.08 \times 10^{-10} \text{ m/s}^2 \quad \leftarrow \text{derived from } S^1 \text{ fiber geometry}$$

Closed-form solution:

```
def a_bst(a_N, a0=3702.6):    # a0 in (km/s)2/kpc
    xN = a_N / a0
    u = (xN**2 + xN * np.sqrt(xN**2 + 4)) / 2
    return np.sqrt(u) * a0
```

Mass-to-light ratios fixed by stellar population synthesis (not fitted): - $Y_{\text{disk}} = 0.5 M_{\odot}/L_{\odot}$ - $Y_{\text{bul}} = 0.7 M_{\odot}/L_{\odot}$

Free parameters: 0

Results Summary

Metric	Value
Galaxies	175 (full SPARC sample)
Median RMS	12.4 km/s
Median RMS%	11.8%
Free parameters	0
Dark non-matter particles required	None

68% of high-quality galaxies ($Q=1$) fit within 15% RMS with zero free parameters — comparable to MOND with the same fixed mass-to-light priors, while BST derives the interpolating function and a_0 from first principles.

SPARC Data Attribution

Rotation curve data from:

Lelli, F., McGaugh, S. S., & Schombert, J. M. (2016). SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves. *The Astronomical Journal*, 152, 157.

Data available at: <http://astroweb.cwru.edu/SPARC/>